

Math Made Clear

Optional Subtitle

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About the Author

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감사의 말 간단한 감사 인사나 크레딧

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Contents

Preface	viii
Introduction	ix
1 Numbers & Operations	1
1.1 Place Value	1
1.1.1 What is Place Value?	2
1.2 Decimals and Fractions	3
1.2.1 What are Decimals and Fractions?	4
1.3 Comparing and Ordering Numbers	4
1.3.1 How do we compare and order numbers?	5
1.4 Factors and Multiples	6
1.4.1 What are Factors and Multiples?	7
1.5 Prime and Composite Numbers	7
1.5.1 What are Prime and Composite Numbers?	8
1.6 Greatest Common Factor & Least Common Multiple	8
1.6.1 What are GCF and LCM?	9
1.7 Tip for Students	9
1.8 Additional Concepts	11
1.8.1 Place Value Vocabulary	11
1.8.2 Fraction Fundamentals	12
1.8.3 Decimal Types	13
1.8.4 Comparison Vocabulary	14
1.8.5 Prime Factorization	14
1.8.6 Finding GCF and LCM	15
2 Variables and Algebra	18
2.1 Variables and Algebraic Expressions	18
2.1.1 What You'll Learn	19

2.1.2	Key Ideas	19
2.1.3	Everyday Example	20
2.1.4	Did You Know?	20
2.1.5	Tips for Students	20
2.2	Simplifying Expressions	20
2.2.1	What You'll Learn	20
2.2.2	Key Ideas	21
2.2.3	Why It Matters	21
2.2.4	Did You Know?	21
2.2.5	Tips for Students	21
2.3	Solving One-Step Equations	21
2.3.1	What You'll Learn	22
2.3.2	Key Ideas	22
2.3.3	Real-World Example	22
2.3.4	Did You Know?	23
2.3.5	Tips for Students	23
2.4	Solving Multi-Step Equations	23
2.4.1	What You'll Learn	23
2.4.2	Key Ideas	23
2.4.3	How This Helps	24
2.4.4	Did You Know?	24
2.4.5	Tips for Students	24
2.5	Inequalities	24
2.5.1	What You'll Learn	25
2.5.2	Key Ideas	25
2.5.3	Everyday Example	25
2.5.4	Did You Know?	26
2.5.5	Tips for Students	26
3	Ratios, Rates, and Percents	27
3.1	Ratios	27
3.1.1	What is a Ratio?	28
3.1.2	Did You Know?	29
3.1.3	Tips for Students	29
3.2	Rates	29
3.2.1	What is a Rate?	30
3.2.2	Did You Know?	31
3.2.3	Tips for Students	31
3.3	Unit Rate	31
3.3.1	What is a Unit Rate?	32
3.3.2	Did You Know?	32
3.3.3	Tips for Students	32

3.4	Proportions	33
3.4.1	What is a Proportion?	33
3.4.2	Did You Know?	34
3.4.3	Tips for Students	34
3.5	Percents	34
3.5.1	What is a Percent?	35
3.5.2	Conversions	35
3.5.3	Real-Life Examples	36
3.5.4	Did You Know?	36
3.5.5	Tips for Students	36
3.6	Tip for Students	36
4	Geometry	37
4.1	Lines, Angles, and Shapes	37
4.1.1	What you'll learn	38
4.1.2	Key vocabulary	38
4.1.3	Short explanation and examples	38
4.1.4	Types of angles (simple list)	39
4.1.5	Try this (brief activity)	39
4.2	Triangles and Quadrilaterals	39
4.2.1	What you'll learn	39
4.2.2	Triangles — quick guide	40
4.2.3	Quadrilaterals — quick guide	40
4.2.4	Short activity	40
4.3	Circles and Parts of a Circle	41
4.3.1	What you'll learn	41
4.3.2	Key parts	41
4.3.3	Formulas (use simple numbers)	41
4.3.4	Example	42
4.4	Area and Perimeter	42
4.4.1	What you'll learn	42
4.4.2	Formulas (memorize the short list)	42
4.4.3	Short practice	43
4.5	Surface Area and Volume	43
4.5.1	What you'll learn	43
4.5.2	Quick formulas	43
4.5.3	Think about it	44
4.6	Coordinate Plane	44
4.6.1	What you'll learn	44
4.6.2	Key ideas	44
4.6.3	Try it	44
4.7	Transformations	45

4.7.1	What you'll learn	45
4.7.2	Short definitions and examples	45
4.7.3	Classroom activity	46
4.8	Tip for Students	46
Appendix		47
References		49

Preface

Mathematics is a universal language, but learning it in a second language can be both a challenge and an opportunity. This book was created especially for Korean students who are studying math in English. Our purpose is to bridge the gap between two worlds—helping you master essential math concepts while also building your confidence in English.

수학은 보편적인 언어이지만, 제2언어로 수학을 배우는 것은 도전이면서 동시에 기회가 될 수 있습니다. 이 책은 영어로 수학을 공부하는 한국 학생들을 위해 특별히 만들어졌습니다. 우리의 목적은 두 세계 사이의 간극을 메우는 것입니다—필수적인 수학 개념을 마스터하면서 동시에 영어에 대한 자신감을 키우는 것을 돕는 것입니다.

The motivation for this book comes from the experiences of many students who have struggled to understand math vocabulary, instructions, and problem-solving strategies in English. We know that with the right support, every student can succeed. By providing clear explanations, bilingual vocabulary, and step-by-step examples, we hope to make math accessible and enjoyable for you.

이 책의 동기는 영어로 된 수학 용어, 지시사항, 문제 해결 전략을 이해하는 데 어려움을 겪은 많은 학생들의 경험에서 비롯됩니다. 우리는 적절한 지원이 있으면 모든 학생이 성공할 수 있다는 것을 알고 있습니다. 명확한 설명, 이중 언어 어휘, 단계별 예시를 제공함으로써, 우리는 수학을 접근하기 쉽고 즐겁게 만들어 드리기를 희망합니다.

Whether you are preparing for international exams, planning to study abroad, or simply want to strengthen your math and language skills, this book is here to support your journey. Remember, every great mathematician started with curiosity and perseverance. We hope this book inspires you to explore, ask questions, and discover the beauty of mathematics in both Korean and English.

국제 시험을 준비하거나, 해외 유학을 계획하거나, 단순히 수학과 언어 능력을 강화하고 싶은 간에, 이 책은 여러분의 여정을 지원하기 위해 여기에 있습니다. 기억하세요, 모든 위대한 수학자는 호기심과 끈기로 시작했습니다. 우리는 이 책이 여러분이 탐구하고, 질문하고, 한국어와 영어로 된 수학의 아름다움을 발견하도록 영감을 주기를 희망합니다.

Introduction

Welcome to this math book designed especially for Korean students learning English! This book is organized to help you build a strong foundation in mathematics while improving your English skills. Each section covers a key topic, starting with clear explanations and followed by examples, practice questions, and helpful tips.

영어를 배우는 한국 학생들을 위해 특별히 설계된 이 수학책에 오신 것을 환영합니다! 이 책은 여러분이 영어 실력을 향상시키면서 동시에 수학에 대한 강한 기초를 쌓을 수 있도록 구성되어 있습니다. 각 섹션은 핵심 주제를 다루며, 명확한 설명으로 시작하여 예시, 연습 문제, 유용한 팁으로 이어집니다.

The book is divided into major sections such as Numbers & Operations, Expressions & Equations, Geometry, Data & Statistics, and more. Within each section, you will find step-by-step guides, vocabulary lists, and real-world applications. Special features include bilingual glossaries, common mistakes to avoid, and mini quizzes to check your understanding.

이 책은 수와 연산, 식과 방정식, 기하학, 데이터와 통계 등의 주요 섹션으로 나뉩니다. 각 섹션 내에서는 단계별 가이드, 어휘 목록, 실제 응용 사례를 찾을 수 있습니다. 특별한 기능으로는 이중 언어 용어집, 피해야 할 일반적인 실수, 이해도를 확인할 수 있는 미니 퀴즈가 포함되어 있습니다.

To get the most out of this book, we recommend reading each section carefully, trying the examples on your own, and using the vocabulary lists to strengthen your math English. Do not hesitate to revisit earlier sections if you need a refresher. Remember, learning math in a new language takes time and practice—be patient with yourself and celebrate your progress!

이 책을 최대한 활용하려면, 각 섹션을 주의 깊게 읽고, 예시를 직접 시도해보며, 어휘 목록을 사용하여 수학 영어를 강화하는 것을 권장합니다. 복습이 필요하면 이전 섹션을 다시 방문하는 것을 주저하지 마세요. 기억하세요, 새로운 언어로 수학을 배우는 것은 시간과 연습이 필요합니다—자신에게 인내심을 가지고 진전을 축하하세요!

We hope this book becomes a valuable companion on your journey to mastering math and English together.

우리는 이 책이 여러분이 수학과 영어를 함께 마스터하는 여정에

서 소중한 동반자가 되기를 희망합니다.

Chapter 1

Numbers & Operations

Welcome! In this book, you'll explore the world of numbers and how we use them every day. You'll learn about place value, decimals, fractions, and more. Each Section includes explanations, examples, and practice problems to help you master these important math skills.

1.1 Place Value

In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number. This idea is called place value. Each place in a number is ten times the value of the one to its right. A number like 4275 is not just four digits. It is a structure: $4000 + 200 + 70 + 5$. We do not read numbers one digit at a time. We read them as a system, where each part works because of its position.

Place value turns a few digits into large or small numbers. Take 9836. The 8 is not just an 8. It is 800, because it sits in hundreds of places. In 5104, the 0 adds nothing but still matters. It shows that there are no tens. In 12345, the 2 sits in the thousands place and means 2000. These examples may seem simple, but they show a key truth: digits mean more when you know where they stand.

“In 5104, the 0 adds nothing but still matters.”

5104에서 0은 자릿수 유지만을 위해 존재한다 십의 자리가 비어 있음을 보여주는 중요한 역할 숫자 위치를 유지해 주는 자리 표시자

Place value is more than vocabulary. It affects how we add, subtract, estimate, and think about numbers. When we know what each digit means, we can break numbers apart and put them back together. We stop guessing and start understanding. That shift helps us see math not as a list of rules but as a system that makes sense.

“In our number system, every digit has a purpose. A 4 does not always mean 4. Sometimes it means 4000. That depends on where it sits in the number.”

자리값은 숫자의 위치에 따라 달라짐 왼쪽으로 한 자리 이동할 때 마다 그 10배 예시: 10의 자리는 1의 자리보다 10배 큼

“Each place in a number is ten times the value of the one to its right.”

자리값은 숫자의 위치에 따라 달라짐 왼쪽으로 한 자리 이동할 때 마다 그 10배 예시: 10의 자리는 1의 자리보다 10배 큼

1.1.1 What is Place Value?

Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.

To truly grasp the concept of place value, for example, let's take a closer look at the number 4275. Here, the digit 4 is found in the thousands place, which means it actually represents 4000, not just 4. Moving to the right, the 2 is in the hundreds place, so it stands for 200. The 7, sitting in the tens place, is worth 70, and finally, the 5 in the ones place simply means 5. This system, where each digit's value increases by a factor of ten as you move left, is what makes our decimal system so powerful and flexible.

To help visualize this, consider the following place value chart:

Place	Value
Thousands	1000
Hundreds	100
Tens	10
Ones	1

Let's see how this works in practice. If we write 4275 in expanded form, we get:

$$4275 = 4000 + 200 + 70 + 5$$

Now, imagine you are looking at the number 9836. You might wonder, what is the value of the digit 8 in this number? Since the 8 is in the hundreds place, its value is 800. This simple observation shows how the position of a digit determines its true value. Similarly, if you break down 5104 into its expanded form, you see it as $5000 + 100 + 0 + 4$, making it clear what each digit stands for. In another example, the digit 2 in 12345 is in the thousands place, so it represents 2000. Understanding these positions helps us see the true value of every digit in a number, which is a fundamental skill in mathematics.

“Place value tells us the value of each digit in a number, depending on its position. In our decimal system, each place is ten times the value of the place to its right.”

자리값은 숫자의 위치에 따라 달라짐 왼쪽으로 한 자리 이동할 때
마다 그 10배 예시: 10의 자리는 1의 자리보다 10배 큼

1.2 Decimals and Fractions

Sometimes, a whole number is too much. We want to describe something in parts: half a sandwich, a third of a mile, seventy-five hundredths of a dollar. That's when we use fractions and decimals. There are two ways of saying the same thing: less than one. Fractions use numerators and denominators. Decimals use place value. But both express parts of a whole.

Decimals and fractions are not separate languages. They translate. 0.75 is the same as $\frac{75}{100}$, which reduces to $\frac{3}{4}$. The fraction $\frac{3}{10}$ becomes 0.3. The decimal 0.6 becomes $\frac{6}{10}$, which is $\frac{3}{5}$. Being able to switch between them is not just a trick. It is a skill. And that skill helps you compare, calculate, and make better sense of the numbers you see every day.

“Decimals and fractions are not separate languages. They translate. 0.75 is the same as $\frac{75}{100}$, which reduces to $\frac{3}{4}$.”

소수랑 분수는 같은거임 예시: $0.25 = 25/100 = 1/4$

1.2.1 What are Decimals and Fractions?

As we move beyond whole numbers, we encounter quantities that are less than one. These are often represented as fractions or decimals. A **fraction** shows a part of a whole, such as $\frac{1}{2}$ or $\frac{3}{4}$. On the other hand, a **decimal** expresses the same idea using a decimal point, like 0.5 or 0.75. Both forms are just different ways to describe the same concept.

For example, consider the decimal 0.75. This can be written as the fraction $\frac{75}{100}$, which simplifies to $\frac{3}{4}$.

$$0.75 = \frac{75}{100} = \frac{3}{4}$$

Let's try converting between these forms. If you have the fraction $\frac{3}{10}$, you can write it as the decimal 0.3. Conversely, if you start with the decimal 0.6, you can express it as a fraction: $0.6 = \frac{6}{10} = \frac{3}{5}$ in simplest form. Being able to move between decimals and fractions is a key skill that will help you in many areas of math.

소수랑 분수는 같은거임 예시: $0.25 = 25/100 = 1/4$

1.3 Comparing and Ordering Numbers

Math is not just about knowing what numbers are. It is about knowing how they relate. We compare. We rank. We ask, which is

bigger? Which is smaller? Which is equal? Symbols like $>$, $<$, and $=$ help us answer. But first, we have to understand the values.

Take 5.03 and 5.3. They may look close. But 5.3 is larger. Now try comparing fractions: $\frac{2}{5}$ and $\frac{3}{7}$. Convert them to decimals. $\frac{2}{5}$ is 0.4. $\frac{3}{7}$ is about 0.428. So $\frac{3}{7}$ is greater. This kind of comparison matters in real life—from prices to measurements to data.

Ordering numbers builds the same skill. Imagine you are given 0.2, $\frac{1}{3}$, 0.25, and 0.199. Which is the greatest? Which is least? Convert, compare, and you find the order. This process teaches more than math. It teaches precision, logic, and clarity.

“Math is not just about knowing what numbers are. It is about knowing how they relate.”

비교, 비교, 질문, 고뇌 비교후 고뇌를 통해 뭐가 큰지 찾으세요

“We compare. We rank. We ask, which is bigger? Which is smaller?”

비교, 비교, 질문, 고뇌 비교후 고뇌를 통해 뭐가 큰지 찾으세요

“Symbols like $>$, $<$, and $=$ help us answer.”

기호 설명 덧붙이기: $<$, $<=$, $>$, $>=$ greater than ($>$), greater than or equal to ($>=$), less than ($<$), less than or equal to ($<=$)

1.3.1 How do we compare and order numbers?

In mathematics, it's important not only to understand numbers, but also to compare them and put them in order. We use symbols like $>$ (greater than), $<$ (less than), and $=$ (equal to) to show how

numbers relate to each other. For example, if you are given two numbers, you can decide which is larger or smaller, or if they are the same.

Let's look at a few examples. Suppose you want to compare 5.03 and 5.3. Since $5.03 < 5.3$, we use the less than symbol. Now, what if you have two fractions, $\frac{2}{5}$ and $\frac{3}{7}$? By converting them to decimals, you find that $\frac{2}{5} = 0.4$ and $\frac{3}{7} \approx 0.428$, so $0.4 < 0.428$, which means $\frac{2}{5} < \frac{3}{7}$.

Ordering numbers is also a useful skill. Imagine you are asked to arrange the numbers 0.2, $\frac{1}{3}$, 0.25, 0.199 from greatest to least. By converting them all to decimals, you see that $\frac{1}{3}$ (approximately 0.333) is the largest, followed by 0.25, then 0.2, and finally 0.199. Sometimes, you may need to compare a fraction and a decimal, such as $\frac{4}{9}$ and 0.44. Since $\frac{4}{9} \approx 0.444$, it is slightly greater than 0.44. These skills are essential for working with numbers in real-life situations.

비교, 비교, 질문, 고뇌 비교후 고뇌를 통해 뭐가 큰지 찾으세요

1.4 Factors and Multiples

Some numbers fit together cleanly. Others do not. Factors and multiples explain why. A factor is a number that divides evenly. A multiple is what you get when you multiply. These two ideas help us break numbers apart and build them back up.

If you list the factors of 12, you get 1, 2, 3, 4, 6, and 12. Each divides 12 without a remainder. The multiples of 5 are 5, 10, 15, 20, and so on. These ideas may seem mechanical, but they are powerful. They lay the groundwork for division, fractions, and algebra.

“A factor is a number that divides evenly. A multiple is what you get when you multiply.”

factor = 약수; 어떤 수나 식을 나누어 나머지가 없이 떨어지는 수
나 식을 일컫는 말 예시: 3은 6의 약수이다; $6/3=2$

multiple = 배수; 어떤 특정 정수에 다른 정수를 곱해서 얻어지는
수 예시: 6은 3의 배수이다; $3*2=6$

1.4.1 What are Factors and Multiples?

As you continue your journey in mathematics, you'll often need to break numbers down or build them up. This is where factors and multiples come in. A **factor** is a number that divides another number exactly, leaving no remainder. For example, the factors of 12 are 1, 2, 3, 4, 6, and 12, because each of these divides 12 evenly. On the other hand, a **multiple** is what you get when you multiply a number by an integer. For instance, the multiples of 5 are 5, 10, 15, 20, and so on.

Let's see how this works with a few examples. If you want to find all the factors of 24, you list out 1, 2, 3, 4, 6, 8, 12, and 24. If you're asked for the first 6 multiples of 8, you simply multiply: 8, 16, 24, 32, 40, and 48. Sometimes, you might wonder if one number is a factor of another. For example, is 9 a factor of 81? Since $81 \div 9 = 9$ with no remainder, the answer is yes. Understanding factors and multiples is crucial for topics like division, fractions, and finding patterns in numbers.

1.5 Prime and Composite Numbers

Some numbers stand alone. Others are made of parts. Prime numbers have only two factors: 1 and themselves. Composite numbers have more. This is not just a label—it is a clue to how numbers work.

2, 3, 5, 7, 11, 13, and 17 are prime. They cannot be broken down. But 15 is composite: it splits into 3 and 5. Learning to recognize prime and composite numbers helps us factor numbers, simplify fractions, and understand the building blocks of math.

“Prime numbers have only two factors: 1 and themselves.
Composite numbers have more.”

prime number = 소수; 어떤 특정 정수에 다른 정수를 곱해서 얻어지는 수 예시: 2, 3, 5, 7, 11, etc

composite number = 합성수; 1보다 큰 자연수 중 소수가 아닌 수 예시: $10(5 \times 2)$, $35(7 \times 5)$, $40(8 \times 5)$

1.5.1 What are Prime and Composite Numbers?

As you explore numbers further, you'll discover that some numbers have only two factors, while others have many. A **prime number** is a number greater than 1 that has exactly two distinct factors: 1 and itself. For example, 2, 3, 5, 7, 11, 13, 17, and 19 are all prime numbers under 20. In contrast, a **composite number** has more than two factors. For instance, 15 is composite because it can be divided by 1, 3, 5, and 15.

Let's practice identifying these. Consider the numbers 14, 17, 21, 23, and 25. Among these, 17 and 23 are prime because they have only two factors. The others—14, 21, and 25—are composite. If you look at all the composite numbers between 10 and 20, you'll find 10, 12, 14, 15, 16, 18, and 20. It's also important to note that the number 1 is neither prime nor composite, since it has only one factor: itself. Recognizing prime and composite numbers is a foundational skill for understanding more advanced math concepts like prime factorization and divisibility.

1.6 Greatest Common Factor & Least Common Multiple

Sometimes two numbers meet in the middle. The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number that both numbers share as a multiple. These ideas sound opposite, but they both help us solve problems with shared values.

Take 12 and 18. Their GCF is 6 since it is the biggest number that divides them both. Their LCM is 36 which is the smallest number they both fit into. We find these using prime factorization. Once you learn how, you see how numbers connect. And once you see the connection, math becomes clearer.

“The Greatest Common Factor (GCF) is the biggest number they both share as a factor. The Least Common Multiple (LCM) is the smallest number that both numbers share as a multiple.”

GCF = 최대공약수; 1보다 큰 자연수 중에서 1과 자기 자신만을 약수로 갖는 소수가 아닌 수 예시: 12와 18의 최대공약수는 6 (나 예시)

LCM = 최소공배수; 두 개 이상의 여러 수에 공통으로 존재하는 배수들 중에서 가장 작은 수 예시: 12와 18의 최소공배수는 36 (나 예시)

1.6.1 What are GCF and LCM?

When working with two or more numbers, it's often useful to find numbers they have in common. The **Greatest Common Factor (GCF)** is the largest number that divides two numbers exactly, while the **Least Common Multiple (LCM)** is the smallest number that both numbers divide into evenly.

To find the GCF and LCM, you can use prime factorization. For example, let's find the GCF and LCM of 12 and 18. The prime factorization of 12 is $2^2 \times 3$, and for 18 it's 2×3^2 . The GCF is the product of the lowest powers of common primes: $2^1 \times 3^1 = 6$. The LCM is the product of the highest powers: $2^2 \times 3^2 = 36$.

Let's apply this to a few more examples. The GCF of 24 and 36 is 12, since 12 is the largest number that divides both. The LCM of 8 and 12 is 24, which is the smallest number both can divide into. For 14 and 28, we see that $14 = 2 \times 7$ and $28 = 2^2 \times 7$, so the GCF is 14 and the LCM is 28. Mastering GCF and LCM will help you with fractions, ratios, and solving many real-world problems.

1.7 Tip for Students

Math is not just about solving problems. It is about seeing how ideas fit together. Learn the basics—place value, fractions, comparisons, factors, primes, and multiples. They are not just chapters. They are tools. And the better you understand them, the more confidently you can use them anywhere.

Interesting Facts

- The word “digit” comes from the Latin word for finger, because people used their fingers to count.
- The decimal system is called “base-10” because each place is ten times the value of the place to its right.

- The largest single-digit number in the decimal system is 9.
- The introduction of zero by Indian mathematicians made calculations much clearer and easier.

1.8 Additional Concepts

This section covers essential vocabulary and techniques that will help you master the fundamental concepts of numbers and their relationships. These additional tools complement the main concepts and provide practical methods for solving problems.

1.8.1 Place Value Vocabulary

Understanding place value requires familiarity with specific terminology. The essential place values you need to learn include billions, hundred millions, ten millions, millions, hundred thousands, ten thousands, thousands, hundreds, tens, and ones.

“한국어와 비슷하게 앞자리에 10배만 붙이면 됨: 일 십 백 천 만 십만 백만 천만 억 십억 백억”

“ones(일), tens(십), hundreds(백), thousands(천), ten thousands(만) 영어는 천부터 $\times 10$ ”

For decimal place values, the key terms are tenths, hundredths, thousandths, ten-thousandths, and millionths.

“똑같은 건데 뒤에 th만 추가”

“반드시 알아야하는 건 tenths(할), hundredths(푼), thousandths(리)”

“이외의 것은 굳이”

1.8.2 Fraction Fundamentals

Fractions represent parts of a whole, and understanding their components is crucial. Every fraction has two main parts: the numerator and the denominator.

“denominator(분모), numerator(분자)”

“외우기 힘들면 그냥 AND로 외우면 됨 and 순서대로 n/d ”

There are different types of fractions that serve different purposes. An improper fraction has a numerator larger than its denominator, while a mixed number combines a whole number with a proper fraction.

“improper(가분수), mixed(대분수)”

“이건 솔직히 그냥 성질만 알면 바로 외워짐; 가분수는 대가리가 크니까 당연히 비율적으로 부적절(improper)함; 대분수는 자연수와 분수의 혼합(mix)”

1.8.3 Decimal Types

Decimals come in different forms, each with unique characteristics. A repeating or recurring decimal has a pattern that continues infinitely, while a finite decimal has a limited number of digits after the decimal point. An infinite decimal continues without end.

“repeating/recurring(순환소수), finite(유한), infinite(무한)”

“repeating/recurring(반복), in-(없다) + finite(제한이 있는) = infinite(무제한)”

“순환소수의 구별방법은 소수점위에 점의 유무 (ex1: 1.3-위에 점 찍혔다 치면 1.333333333;”

“ex2: 1.3475 3과 5위에 점이 찍혔을 경우 그 구간이 순환 1.347534753475)”

“무한소수는 분수로의 변환이 불가하다”

“순환소수의 분수 변환은 변환구간을 9로 나눈다 (ex:

1.3984에서 3984가 순환할 경우 $3984/9999 = 0.398439843984 \dots$)”

1.8.4 Comparison Vocabulary

Mathematics uses specific language to describe relationships between numbers. Understanding these terms helps you interpret problems correctly and communicate mathematical ideas clearly.

For greater than relationships, we use terms like greater than, more than, larger than, longer than, higher than, above, exceeds, and over. For greater than or equal to, we say greater than or equal to, at least, more than or equal to, minimum, no less than, not under/below, or not smaller than.

For less than relationships, we use less than, fewer than, smaller than, shorter than, lower than, under, below, or beneath. For less than or equal to, we say at most, does not exceed, maximum, no more than, up to, or not above/greater than.

“좀 많긴 한데 at least/minimum 이랑 at most/maximum 만
외우면 sat 수학은 해결”

“무작정 외우지 말고 문제 읽을때 한번 뜻을 생각해 보
면 이해가 됨”

1.8.5 Prime Factorization

Prime factorization is the process of breaking down a number into its smallest building blocks—prime factors. This technique is fundamental for understanding number structure and solving problems involving factors and multiples.

“prime factorization(소인수분해) 말그대로 prime(소수)까지 분해한다는 거임”

“제일 기본적인 순서는 소인수분해할 숫자를 보는거임”

“그 숫자가 몇의 배수인지 봐야하는데 이걸”

To determine if a number is divisible by certain factors, we can use these rules:

- 2의 배수: 짝수/일의 자리 숫자가 2, 4, 6, 8, 0인 수
- 3의 배수: 각 자리의 수의 합이 3의 배수
- 끝의 두 자리 수가 00이거나 4의 배수
- 5의 배수: 일의 자리 숫자가 0또는 5
- 7의 배수: 세자리수 abc일 때 $ab-c \times 2$ 가 7의 배수인수 (이건 솔직히 몰라도 됨)

“일단 대충 저걸로 판단후 최소 값부터 계속 소인수분해 (ex: 60을 소인수 분해 하면 일단 2로 나눈다 왜냐 하면 짝수이기 때문; 2×30 ; 30도 짝수이기 때문에 2로 다시 나눈다; $2 \times 2 \times 15$; 15는 일의 자리가 5이기 때문에 5로 나눈다; $2 \times 2 \times 5 \times 3$; 와 다 소수네; 소인수분해 완료)”

1.8.6 Finding GCF and LCM

The greatest common factor (GCF) and least common multiple (LCM) are essential concepts for working with multiple numbers.

While these ideas may seem complex, there's a simple method using the Korean "ㄴ" technique that makes finding them straightforward.

“한국 학교에서 당연히 배웠겠지만 'ㄴ'자로 하면됨”

“'ㄴ'자 방법이란 소인수분해랑 똑같은건데 두수 혹은 그 이상의 숫자들이 공유하는 약수로 다 나눈건데 이거는 꼭 소수로 안해도 됨”

Let's find the GCF of 12 and 18 using this method:

$$\begin{array}{r|rr} 6 & 12 & 18 \\ \hline & 2 & 3 \end{array}$$

“위에처럼 마지막 숫자가 소수가 되면 끝”

“최대공약수의 경우 'ㄴ'자 옆에 있는 숫자들의 곱; 지금 같은 경우는 6”

Now let's find the LCM of 12 and 18:

$$\begin{array}{r|rr} 6 & 12 & 18 \\ \hline & 2 & 3 \end{array}$$

“최소공배수의 경우 'ㄴ'자 옆 밑에 있는 모든 숫자들의
곱; 지금 같은 경우는 $6*2*3 = 36$ ”

“와 쉽죠”

These methods provide efficient ways to work with factors and multiples, making complex problems more manageable and helping you develop a deeper understanding of number relationships.

Chapter 2

Variables and Algebra

(문자와 대수)

Welcome to the world of algebra! In this chapter, we'll learn how to use letters like **x** and **y** to represent numbers. These letters are called **variables**, and they help us write expressions and solve equations. Algebra might seem like a new language, but once you understand the rules, it can actually make math easier. Let's explore how we can describe patterns, solve problems, and think logically using simple algebraic tools.

Variables are like mystery boxes that can hold different numbers. When we write expressions like $3x + 5$, we're creating a rule that works for any value of x . This powerful idea helps us solve problems where we don't know the answer yet, but we know how things relate to each other.

2.1 Variables and Algebraic Expressions

(문자와 식)

Welcome to the world of algebra! In this chapter, we'll learn how to use letters like **x** and **y** to represent numbers. These letters are called **variables**, and they help us write expressions and solve equations. Algebra might seem like a new language, but once you understand the rules, it can actually make math easier. Let's explore how we can describe patterns, solve problems, and think logically

using simple algebraic tools.

“A variable is a symbol, like x or y , that stands for a number.”

variable (x , y , z , etc) 는 보통 문자로 표현되고, 어떤 수를 대신 나타냄
예시: $x = 8$; $7x = 56$

2.1.1 What You’ll Learn

- Understand what variables are and why we use them
- Learn how to write and read algebraic expressions

2.1.2 Key Ideas

- A **variable** is a symbol, like x or y , that stands for a number.
- An **algebraic expression** is a mix of numbers, variables, and math operations such as $+$, $-$, $\times$, or $\div$.

“An expression is a mix of numbers, variables, and operations.”

expression(식)은 숫자, 문자(변수), 그리고 연산($+$, $-$, $\times$, $\div$)이 섞여 있는 수학적 표현
예시: $3x + 5$, $7y^2$

Example:

$3x + 5$ means “3 times some number, then add 5.”

If $x = 2$, then:

$$3x + 5 = 3(2) + 5 = 6 + 5 = 11$$

2.1.3 Everyday Example

Imagine you earn \$10 per hour. If h is the number of hours you work, then your earnings are written as:

$$10h$$

If you work 5 hours:

$$10(5) = 50$$

2.1.4 Did You Know?

The word **algebra** comes from the Arabic term *al-jabr*, which means “reunion of broken parts.” It appeared in a famous math book written over 1,200 years ago by a scholar named Al-Khwarizmi. Many English math words like *algorithm* also come from his name!

2.1.5 Tips for Students

- Think of a variable as a mystery box.
- Try different values to see how expressions change.
- Use colors, arrows, or circles to match terms and operations.

2.2 Simplifying Expressions

(식의 간단화)

“Like terms have the same variable and exponent.”

like terms(동류항)은 같은 문자를 포함하는 항 예시: $2x$, $3x$ 는 동류항이며, 합쳐서 $5x$

2.2.1 What You’ll Learn

- Combine like terms
- Use the distributive property to simplify expressions

2.2.2 Key Ideas

- **Like terms** have the same variable and exponent.
- The **distributive property**:

$$a(b + c) = ab + ac$$

“The distributive property is another tool: $3(x + 4)$ is the same as $3x + 12$.”

distributive property(분배법칙)은 괄호를 풀 때 사용 예: $3(x + 4) = 3x + 12$

Examples:

$$2x + 3x = 5x$$

$$3(x + 4) = 3x + 12$$

2.2.3 Why It Matters

Simplifying expressions helps make your work cleaner and your thinking clearer.

2.2.4 Did You Know?

Even NASA engineers and video game developers simplify expressions when building models and writing code.

2.2.5 Tips for Students

- Circle or highlight like terms before combining.
- Be careful with minus signs.
- Go one step at a time.

2.3 Solving One-Step Equations

(1단계 방정식)

“Use inverse operations to isolate the variable. Do the same to both sides.”

“Equations are like balances. Whatever you do to one side, you must do to the other.”

equation(방정식)은 balance(저울)과 같습니다. 한쪽에 한 연산을 하면, 반대쪽도 똑같이 해야 균형이 유지됨

2.3.1 What You’ll Learn

- Solve equations in one step
- Understand inverse operations

2.3.2 Key Ideas

- Use **inverse operations** to isolate the variable.
- Do the same thing to both sides of the equation.

“A one-step equation is solved with just one move—addition, subtraction, multiplication, or division.”

one-step equation(1단계 방정식)은 한 번의 연산으로 해결되는 식
예시: $x + 7 = 12 \rightarrow x = 5$

Examples:

$$x + 4 = 9 \Rightarrow x = 5$$

$$5x = 20 \Rightarrow x = 4$$

2.3.3 Real-World Example

If \$20 is 5 times the cost of one notebook:

$$5x = 20 \Rightarrow x = 4$$

2.3.4 Did You Know?

Solving equations is like solving a riddle or puzzle.

2.3.5 Tips for Students

- Ask: “What’s happening to the variable?” Then do the opposite.
- Keep both sides equal.
- Plug your answer back in to check.

2.4 Solving Multi-Step Equations

(여러 단계 방정식)

“First, simplify both sides. Then, isolate the variable step by step.”

“Not all equations are simple. Some take more than one step.”

모든 방정식이 쉽지는 않음. 어떤 것은 여러 단계를 거쳐야만 풀 수 있음

2.4.1 What You’ll Learn

- Solve equations with more than one operation
- Use combining like terms and distribution

2.4.2 Key Ideas

- First, simplify both sides.
- Then, isolate the variable step by step.

“A multi-step equation needs distribution, combining like terms, or more than one operation.”

multi-step equation(여러 단계 방정식)은 분배법칙, 동류항 정리, 또는 여러 연산을 차례대로 사용해야 풀림 예시: $-3x-4(4x-8) = 3(-8x-1) \rightarrow -3x-16x+32 = -24x-3 \rightarrow -19x+32 = -24x-3 \rightarrow -19x+24x+32 = -24x+24x-3 \rightarrow 5x+32 = -3 \rightarrow 5x+32-32 = -3-32 \rightarrow 5x = -35 \rightarrow 5x/5 = -35/5 \rightarrow x = -7$

Example:

$$2(x + 3) = 14$$

$$2x + 6 = 14$$

$$2x = 8$$

$$x = 4$$

2.4.3 How This Helps

Solving multi-step equations builds confidence and logical thinking.

2.4.4 Did You Know?

Multi-step equations are used in budgeting, engineering, and cooking.

2.4.5 Tips for Students

- Don't rush—write each step clearly.
- Simplify both sides before solving.
- Check your solution.

2.5 Inequalities

(부등식)

“If you multiply or divide by a negative number, flip the inequality sign.”

“Inequalities tell us ranges. Instead of saying $x = 5$, we might say $x > 5$.”

inequality(부등식)은 값의 범위를 나타냄 예시: $x = 5$ 대신 $x > 5$ 라고 하면, x 는 5보다 큰 모든 값을 가질 수 있음

2.5.1 What You’ll Learn

- Solve basic inequalities
- Graph answers on a number line

2.5.2 Key Ideas

- Inequality signs:
 - $>$ greater than
 - $<$ less than
 - $\geq$ greater than or equal to
 - $\leq$ less than or equal to
- If you multiply or divide by a negative number, **flip the inequality sign**.

“If you multiply or divide by a negative number, you flip the sign.”

부등식에서 음수로 곱하거나 나눌 때는 부등호 방향을 바꿔야 함
예: $2x < 6 \rightarrow x > 3$

Examples:

$$x - 3 > 2 \Rightarrow x > 5$$

$$-2x < 6 \Rightarrow x > -3$$

2.5.3 Everyday Example

A sign says: “You must be under 18 to enter.” That means:

$$x < 18$$

2.5.4 Did You Know?

Inequalities are used in business, programming, and transportation.

2.5.5 Tips for Students

- Treat inequalities like equations—unless multiplying/dividing by a negative.
- Graph solutions:
 - Open circles ($\circ$) for $<$ or $>$
 - Closed circles ($\bullet$) for $\leq$ or $\geq$
- Test a number to check your solution.

Chapter 3

Ratios, Rates, and Percents

(비율, 속도, 백분율)

Welcome to Unit 3! In this chapter, you'll learn how to compare amounts using *ratios*, measure how fast things change with *rates*, and work with *percents* in real-life situations. These ideas are not just math—they show up in shopping, cooking, sports, and money. Ratios, rates, and percents help us describe relationships between numbers in clear and simple ways.

Ratios help us compare two quantities, like the number of boys to girls in a class or the ingredients in a recipe. Rates measure how things change over time or distance, such as speed in miles per hour or price per item. Percents give us a way to express parts of a whole, making it easy to understand discounts, grades, and statistics.

3.1 Ratios

A ratio is about comparison. It tells us how much of one thing there is compared to another. Ratios are everywhere: the number of players on a team, the ingredients in a recipe, or the colors mixed for paint. A ratio doesn't just give numbers—it shows balance.

If you have 2 apples and 3 oranges, the ratio of apples to oranges is 2 to 3. You can write it $2 : 3$, $\frac{2}{3}$, or “2 to 3.” They all mean the

same thing.

Ratios also show part-to-whole relationships. If a class has 12 boys and 8 girls, the ratio of boys to girls is $12 : 8$, which simplifies to $3 : 2$. But the ratio of girls to total students (20) is $8 : 20$, or $2 : 5$. Ratios let us see relationships clearly.

“Ratios also show part-to-whole relationships.”

비는 부분과 전체의 관계를 나타낼 수도 있음 예시: 8명의 여학생
/ 전체 20명 = 8:20

“A ratio compares two numbers or quantities. It tells us how much of one thing there is compared to another.”

ratio(비)는 두 수나 양을 비교하는 방법 예시: 사과 2개와 오렌지 3개의 비는 2:3

3.1.1 What is a Ratio?

A **ratio** compares two numbers or quantities. It tells us how much of one thing there is compared to another. Ratios help us describe relationships and patterns, and they appear in recipes, sports, and even classrooms.

Examples:

- **Part-to-Part:** apples to oranges $\rightarrow 2 : 3$.
- **Part-to-Whole:** apples to total fruit $\rightarrow 2 : 5$.

Ways to write ratios:

- With a colon: **2 : 3**
- As a fraction: $\frac{2}{3}$
- In words: **2 to 3**

All three ways mean the same thing.

Part-to-Part and Part-to-Whole:

- **Part-to-Part:** Compares two individual parts (e.g., apples to oranges $\rightarrow 2 : 3$).
- **Part-to-Whole:** Compares one part to the total (e.g., apples to total fruits $\rightarrow 2 : 5$).

Example: If a class has 12 boys and 8 girls:

- The ratio of boys to girls is **12 : 8** $\rightarrow$ simplified to **3 : 2**.
- The ratio of girls to total students (20) is **8 : 20** $\rightarrow$ simplified to **2 : 5**.

Ratios simplify comparisons and help us see patterns.

3.1.2 Did You Know?

- Ratios are used to mix paint colors, create scale models, adjust ingredients in recipes, and describe odds in games.
- Ratios are common in recipes, sports, and many real-world mixtures.

3.1.3 Tips for Students

- Keep the order of comparison the same—read carefully.
- Simplify ratios when possible, just like you simplify fractions.
- Think visually—use pictures or charts if it helps you understand.

3.2 Rates

A rate is a special kind of ratio. It compares different units. We use rates to measure speed, prices, and even how fast we type. A car's speed is miles per hour. A store's price is dollars per item. A printer's speed is pages per minute.

If you drive 100 kilometers in 2 hours, the rate is

$$\frac{100}{2} = 50$$

kilometers per hour. Rates make comparisons fair because they put everything on the same scale.

“Rates make comparisons fair because they put everything on the same scale.”

율은 모든 비교를 같은 기준으로 만들어 줌 예: 가격을 비교할 때 \$/개 단위로 계산하면 어떤 상품이 더 저렴한지 알 수 있음

“A rate is a special type of ratio that compares two different kinds of units.”

rate(율)은 특별한 종류의 비로, 서로 다른 단위를 비교함 예시:
 $100\text{km} \div 2\text{시간} = \text{시속 } 50\text{km}$

3.2.1 What is a Rate?

A **rate** is a special type of ratio that compares two different kinds of units. For example, it can compare miles to hours, dollars to items, or words to minutes.

Examples:

- 60 pages in 3 minutes $\rightarrow$ 20 pages per minute.
- \$200 earned in 8 hours $\rightarrow$ \$25 per hour.
- If you drive 100 kilometers in 2 hours, the rate is $\frac{100}{2} = 50 \text{ km/h}$.

Common rates:

- **Speed:** miles per hour (mph), kilometers per hour (km/h).
- **Prices:** dollars per item (\$/item).
- **Typing:** words per minute (wpm).

Everyday Examples:

- A printer prints 60 pages in 3 minutes $\rightarrow$ Rate: **20** pages per minute.
- A worker earns \$200 for 8 hours $\rightarrow$ Rate: **\$25** per hour.

Rates help us measure and compare in real life.

3.2.2 Did You Know?

Rates are everywhere—in recipes, gas mileage (MPG), data usage on your phone, and how fast athletes run.

3.2.3 Tips for Students

- Always include the units (like miles, hours, or dollars).
- To find a rate, divide the first number by the second.
- Use rates to compare which deal is better when shopping.

3.3 Unit Rate

Sometimes we want to know the amount “per one.” That’s called a unit rate. If 3 notebooks cost \$6, then each notebook costs \$2. Unit rates are powerful because they help us compare fairly—per item, per mile, per hour.

“Unit rates are powerful because they help us compare fairly—per item, per mile, per hour.”

단위율은 공정한 비교를 가능하게 해 줌 예시: 사과 가격을 개당으로, 거리 속도를 1시간당으로 비교

When grocery shopping, you might see two boxes of cereal: one \$4 for 8 ounces, another \$6 for 12 ounces. Unit rate shows which is the better deal.

“A unit rate is a rate where the second number is 1. It tells you how much of something there is per one unit of something else.”

unit rate(단위율)은 두 번째 수가 1이 되도록 계산한 율 예시: 노트 3권이 6L, Γ@2 (3:6 → 1:2)

3.3.1 What is a Unit Rate?

A **unit rate** is a rate where the second number is 1. It tells you how much of something there is *per one unit* of something else.

Examples:

- 120 miles in 4 hours $\rightarrow$ 30 miles per hour.
- \$4 for 8 apples $\rightarrow$ \$0.50 per apple.
- If 3 notebooks cost \$6, the unit rate is $\frac{6}{3} = \$2$ per notebook.

How to find a unit rate:

1. Divide the first number by the second.
2. Make sure the second number becomes 1.

Example:

120 miles in 4 hours $\rightarrow 120 \div 4 = 30$ miles per hour.

8 apples for \$4 $\rightarrow \$4 \div 8 = \0.50 per apple.

Real-life uses:

- Grocery stores list unit prices on shelf tags to help shoppers compare.
- Runners track their pace (minutes per mile) to measure speed.
- Restaurants calculate price per serving to set menu prices.

Unit rates appear in budgets, shopping, sports, and cooking.

3.3.2 Did You Know?

Unit prices in stores are unit rates—they show cost per ounce or per item.

3.3.3 Tips for Students

- Make sure the second number becomes 1 (that’s what “unit” means).
- Use real-world examples when practicing—like prices, distance, or time.

- Always include the unit in your answer (like “per hour” or “per item”).

3.4 Proportions

Proportions show equality between two ratios. They tell us that two comparisons are balanced. If a recipe for 2 people uses 1 cup of rice, then a recipe for 4 people uses 2 cups. That’s a proportion: the ratio stays the same.

We solve proportions by cross multiplying. If one ratio equals another, their cross products match. This makes it easy to find missing values.

“We solve proportions by cross multiplying.”

비례식은 cross multiplication(교차 곱셈)으로 풀 수 있음 예: $x/5 = 6/10 \rightarrow 10x = 30 \rightarrow x = 3$

“A proportion is a statement that shows two ratios or rates are equal. It helps us solve problems where quantities grow or shrink at the same rate.”

proportion(비례식)은 두 비가 같음을 나타냄 예: $2/3 = 4/6$

3.4.1 What is a Proportion?

A **proportion** is a statement that shows two ratios or rates are *equal*. It helps us solve problems where quantities grow or shrink at the same rate.

Example:

$$\frac{2}{3} = \frac{4}{6}$$

If $\frac{2}{3} = \frac{4}{6}$, this is a proportion because both sides are equal.

How to solve proportions: Use *cross multiplication*:

$$\frac{2}{3} = \frac{4}{6} \implies 2 \times 6 = 3 \times 4 \implies 12 = 12.$$

This means it is a true proportion.

You can also use proportions to find unknown numbers. For example,

$$\frac{x}{5} = \frac{6}{10} \implies 10x = 30 \implies x = 3.$$

Everyday examples:

- A map says 1 inch = 50 miles $\rightarrow$ use this ratio to calculate real distances.
- A recipe that serves 2 people can be doubled to serve 4 using proportions.

Proportions keep comparisons fair and balanced.

3.4.2 Did You Know?

Proportions are used in architecture, photography, and fashion design to maintain balance and scale.

3.4.3 Tips for Students

- Always simplify ratios before comparing.
- Use cross multiplication to solve quickly.
- Write word problems as proportions to organize your thinking.

3.5 Percents

A percent means “per hundred.” It is another way to compare numbers. 50% is half, 25% is a quarter, and 100% is the whole. Percents are part of everyday life—on price tags, test scores, and bank interest.

Converting between fractions, decimals, and percents is simple once you know the rules. Fractions turn into percents by multiplying by 100. Decimals turn into percents by moving the decimal two places to the right. Percents turn back into decimals by moving the decimal two places to the left.

“Fractions turn into percentages by multiplying by 100.”

분수를 백분율로 바꾸려면 100을 곱하면 됨 예: $1/4 = 0.25 \rightarrow 25\%$

“A percent shows part of a whole, out of 100. It is a number that means “per 100.””

percent(백분율)은 '100분의 '를 의미 예: $50\% = 50/100 = 1/2$

3.5.1 What is a Percent?

A **percent** shows part of a whole, out of 100. It is a number that means “*per 100*.” It shows a part out of a whole divided into 100 equal parts.

Examples:

- 50% means 50 out of 100 $\rightarrow \frac{1}{2}$.
- 25% means 25 out of 100 $\rightarrow \frac{1}{4}$.
- 100% means all of something.

3.5.2 Conversions

- Fraction $\rightarrow$ Percent: $\frac{1}{4} = 0.25 = 25\%$.
- Decimal $\rightarrow$ Percent: $0.6 = 60\%$.
- Percent $\rightarrow$ Decimal: $45\% = 0.45$.

How to convert:

- **Fraction to Percent:** multiply by 100.
 $\frac{1}{4} = 0.25 \Rightarrow 25\%$.
- **Decimal to Percent:** move the decimal point two places to the right.
 $0.6 \Rightarrow 60\%$.

- **Percent to Decimal:** move the decimal point two places to the left.
 $45\% \Rightarrow 0.45$.

3.5.3 Real-Life Examples

- **Shopping:** 30% off a jacket means you save 30 out of every 100 dollars.
- **Grades:** A score of 85% means you got 85 out of 100 points.
- **Interest Rates:** Banks use percents to calculate how much money you earn or owe.

Percents help us see proportions quickly and clearly.

3.5.4 Did You Know?

The percent sign (%) was first used in Europe hundreds of years ago. Today, percents are used globally in business, science, and sports statistics.

3.5.5 Tips for Students

- Think of % as “out of 100.”
- Use visual aids like 100-grids or pie charts to understand percent.
- Practice converting between fractions, decimals, and percents until it feels natural.

3.6 Tip for Students

Ratios, rates, proportions, and percents are not just classroom topics. They describe how things relate and change. Learn to simplify ratios, find unit rates, solve proportions, and convert percents. These tools will help you shop smarter, plan better, and understand the world with clarity.

Chapter 4

Geometry

(기하)

This chapter introduces the basic ideas of geometry in a clear, friendly way. You will learn about points, lines, line segments, and rays; types of angles and how to measure them; simple 2D shapes (triangles, quadrilaterals, circles) and their properties; area, perimeter, surface area, and volume; and the coordinate plane and simple transformations.

Geometry is the study of shapes, sizes, and positions in space. It helps us understand the world around us, from the design of buildings to the patterns in nature. Each section uses short paragraphs and lists to make the ideas easy to read and practice. You'll discover how to measure, classify, and transform shapes using simple tools and clear rules.

4.1 Lines, Angles, and Shapes

(선, 각, 도형)

Geometry begins with the simplest building blocks: points, lines, and angles. A point marks a position in space. A line extends infinitely in both directions. A line segment has two endpoints, while a ray starts at one point and goes on forever in one direction.

An angle forms when two rays meet at a common endpoint called the vertex. We measure angles in degrees, from 0° to 360° . The

most common angles are right angles (90°), which form perfect corners, and straight angles (180°), which form straight lines.

“A point marks a position, a line extends forever, and an angle forms when two rays meet at a vertex.”

4.1.1 What you’ll learn

You will learn what a point, line, line segment, and ray are. You will also meet four common types of angles and see basic flat (2D) shapes.

4.1.2 Key vocabulary

- **Point** (점) — a tiny dot that shows a position.
- **Line** (선) — extends forever in both directions.
- **Line segment** (선분) — part of a line with two endpoints.
- **Ray** (반직선) — starts at one point and goes on forever in one direction.
- **Angle** (각) — made when two rays meet at a point (the vertex).
- **Vertex** (꼭짓점) — the meeting point of two sides or rays.
- **Polygon** (다각형) — a shape made of straight sides (triangle, square, etc.).

4.1.3 Short explanation and examples

A *point* is a place on paper you can mark with a dot. A *line* keeps going and never stops; draw arrows on both ends to show this. A *segment* has two ends; draw a short straight line with dots at both ends. A *ray* has one end and one arrow.

An *angle* is the space between two rays that share the same endpoint. The endpoint is the *vertex*. Use a protractor to measure angles.

4.1.4 Types of angles (simple list)

- **Acute** (예각): less than 90° .
- **Right** (직각): exactly 90° .
- **Obtuse** (둔각): more than 90° and less than 180° .
- **Straight** (평각): exactly 180° (a straight line).

4.1.5 Try this (brief activity)

1. Draw a point and label it A .
2. From A , draw two rays that form a right angle. Label the rays AB and AC .
3. Use a protractor to check that the angle $\angle BAC = 90^\circ$.

4.2 Triangles and Quadrilaterals

(삼각형과 사각형)

Triangles and quadrilaterals are the most common polygons. Triangles have three sides and three angles, while quadrilaterals have four sides and four angles. Each type has special properties that make them useful in geometry and real-world applications.

Triangles can be classified by their sides (equilateral, isosceles, scalene) or by their angles (acute, right, obtuse). Quadrilaterals include familiar shapes like squares, rectangles, and parallelograms, each with their own unique characteristics.

“Triangles have three sides and three angles, while quadrilaterals have four sides and four angles.”

4.2.1 What you'll learn

You will learn how to classify triangles by their sides and angles, and you will meet common four-sided shapes (quadrilaterals).

4.2.2 Triangles — quick guide

Triangles have three sides and three angles. Use the lists below to name a triangle:

By sides

- **Equilateral** (정삼각형): three equal sides and three equal angles.
- **Isosceles** (이등변삼각형): two equal sides.
- **Scalene** (부등변삼각형): no equal sides.

By angles

- **Acute triangle**: all angles $< 90^\circ$.
- **Right triangle**: one angle $= 90^\circ$.
- **Obtuse triangle**: one angle $> 90^\circ$.

4.2.3 Quadrilaterals — quick guide

Quadrilaterals have four sides. Common examples:

- **Square** (정사각형): four equal sides and four right angles.
- **Rectangle** (직사각형): opposite sides equal, four right angles.
- **Rhombus** (마름모): four equal sides (angles may not be right).
- **Parallelogram** (평행사변형): both pairs of opposite sides are parallel.
- **Trapezoid** (사다리꼴): at least one pair of parallel sides.

4.2.4 Short activity

1. Draw one example of each quadrilateral on grid paper.
2. For each, list which sides are equal and which angles look equal.

4.3 Circles and Parts of a Circle

(원과 원의 부분)

Circles are perfectly round shapes where every point on the edge is the same distance from the center. The radius is the distance from the center to any point on the circle, while the diameter is twice the radius and passes through the center.

The circumference is the distance around the circle, and the area is the space inside. These measurements follow simple formulas using the mathematical constant π (pi), which is approximately 3.14.

“A circle is perfectly round with every point on the edge the same distance from the center.”

4.3.1 What you'll learn

You will learn the names of circle parts and two simple formulas: how to find the circumference (distance around) and the area (space inside).

4.3.2 Key parts

- **Center** (중심): the middle point of a circle.
- **Radius** (반지름): distance from center to edge.
- **Diameter** (지름): a straight line through the center connecting two points on the circle. $d = 2r$.
- **Chord** (현): any line segment joining two points on the circle.
- **Circumference** (둘레): distance around the circle.
- **Area**: space inside the circle.

4.3.3 Formulas (use simple numbers)

$$\text{Circumference} = 2\pi r \quad (\text{or } \pi d), \quad \text{Area} = \pi r^2.$$

(Use $\pi \approx 3.14$ for easy calculations.)

4.3.4 Example

If a circle has radius $r = 7$ cm, then

- Circumference $= 2\pi r \approx 2 \times 3.14 \times 7 = 43.96$ cm.
- Area $= \pi r^2 \approx 3.14 \times 49 = 153.86$ cm².

4.4 Area and Perimeter

(넓이와 둘레)

Area and perimeter are two fundamental measurements of shapes. Area measures the space inside a shape, while perimeter measures the distance around the outside. These measurements help us understand how much space shapes take up and how much material we need to build or cover them.

Different shapes have different formulas for calculating area and perimeter. Some shapes, like rectangles and triangles, have simple formulas that are easy to remember and use.

“Area measures the space inside a shape, while perimeter measures the distance around the outside.”

4.4.1 What you’ll learn

Find the area (space inside) and the perimeter (distance around) for common flat shapes.

4.4.2 Formulas (memorize the short list)

- **Rectangle:** Area = length $\times$ width.
Perimeter = $2(\text{length} + \text{width})$.
- **Triangle:** Area = $\frac{1}{2} \times \text{base} \times \text{height}$.
- **Circle:** Area = πr^2 , Circumference = $2\pi r$.

4.4.3 Short practice

1. A rectangle is 8 m long and 3 m wide. Find the area and perimeter.
2. A triangle has base 6 cm and height 4 cm. Find its area.

4.5 Surface Area and Volume

(겉넓이와 부피)

Surface area and volume are measurements of three-dimensional objects. Surface area measures the total area of all the faces or surfaces of an object, while volume measures how much space the object takes up inside.

These measurements are important in real-world applications like packaging, construction, and manufacturing. Understanding the difference helps us choose the right materials and understand how objects relate to space.

“Surface area measures the total area of all faces, while volume measures how much space an object takes up inside.”

4.5.1 What you'll learn

Learn the difference between surface area (sum of all outside faces) and volume (how much fits inside).

4.5.2 Quick formulas

- **Cube** with side s : $SA = 6s^2$, $\text{Volume} = s^3$.
- **Rectangular prism** $l \times w \times h$: $SA = 2(lw + lh + wh)$, $\text{Volume} = lwh$.
- **Cylinder** radius r , height h : $SA = 2\pi r^2 + 2\pi rh$, $\text{Volume} = \pi r^2 h$.

4.5.3 Think about it

Surface area is like the wrapping paper needed to cover a gift box.
Volume is how many small cubes ($1 \times 1 \times 1$) fit inside the box.

4.6 Coordinate Plane

(좌표 평면)

The coordinate plane is a grid system that helps us locate points and describe shapes mathematically. It consists of two perpendicular number lines: the x-axis (horizontal) and the y-axis (vertical). The point where they intersect is called the origin.

Every point on the plane can be described by an ordered pair (x, y) , where x tells us how far to move horizontally from the origin, and y tells us how far to move vertically.

“The coordinate plane uses two perpendicular axes to locate points with ordered pairs (x, y) .”

4.6.1 What you’ll learn

You will place points using ordered pairs (x, y) and draw shapes on the grid.

4.6.2 Key ideas

- The horizontal line is the **x-axis**; the vertical line is the **y-axis**.
- The axes meet at the **origin** $(0, 0)$.
- Read an ordered pair as: move x (left/right), then move y (up/down).

4.6.3 Try it

1. Plot these points: $(1, 1)$, $(4, 1)$, $(4, 3)$, $(1, 3)$.
2. Connect the points in order. What shape did you make?

4.7 Transformations

(이동, 회전, 대칭, 확대/축소)

Transformations are ways to move, rotate, flip, or resize shapes without changing their basic properties. They help us understand how shapes relate to each other and are used in computer graphics, engineering, and many real-world applications.

The four basic transformations are translation (sliding), rotation (turning), reflection (flipping), and dilation (scaling). Each transformation preserves certain properties of the original shape while changing others.

“Transformations move, rotate, flip, or resize shapes while preserving their basic properties.”

4.7.1 What you'll learn

You will learn four simple moves that change where a shape sits or how big it is: translation, rotation, reflection, and dilation.

4.7.2 Short definitions and examples

- **Translation (이동)**: slide the whole shape; it does not turn or change size.
Example: move every point of a triangle 3 units right and 2 units up.
- **Rotation (회전)**: turn the shape around a point (often the origin).
Example: rotate a square 90° clockwise.
- **Reflection (대칭)**: flip the shape across a line (like a mirror).
Example: reflect the point $(2, 1)$ across the y -axis to get $(-2, 1)$.
- **Dilation (확대/축소)**: make the shape bigger or smaller from a center point using a scale factor.

Example: a dilation with factor 2 makes all distances from the center twice as long.

4.7.3 Classroom activity

1. On graph paper, draw a small shape and then:
 - translate it (slide),
 - rotate it 90° ,
 - reflect it across the x -axis,
 - dilate it by factor 1.5.
2. Write one sentence about what changed and what stayed the same after each move.

4.8 Tip for Students

Geometry is not just about memorizing formulas. It is about understanding how shapes work and how they relate to the world around us. Learn to visualize shapes, understand their properties, and see how measurements connect to real problems. These skills will help you in art, architecture, engineering, and many other fields.

Appendix

Important Formulas and Definitions

- **Place Value:** Value of a digit = digit $\times$ place value.
- **Fraction to Decimal:** $\frac{a}{b} = a \div b$.
- **Prime Number:** A number greater than 1 with exactly two factors: 1 and itself.
- **Composite Number:** A number with more than two factors.
- **GCF:** Product of lowest powers of common primes.
- **LCM:** Product of highest powers of all primes present.

Further Reading and Resources

- "Elementary Mathematics" by Liping Ma
- "How to Be a Math Genius" by Mike Goldsmith
- Khan Academy: <https://www.khanacademy.org/>
- Purplemath: <https://www.purplemath.com/>

Summary of Key Concepts

Topic	Key Points
Place Value	Each digit's value depends on its position. Example: $4275 = 4000 + 200 + 70 + 5$.
Decimals and Fractions	Decimals and fractions both represent parts of a whole. Example: $0.75 = \frac{3}{4}$.
Comparing and Ordering Numbers	Use $>$, $<$, $=$ to compare. Convert to decimals for easy comparison. Example: $\frac{2}{5} < \frac{3}{7}$.
Factors and Multiples	Factors divide a number evenly; multiples are products of a number and an integer. Example: Factors of 12 are 1, 2, 3, 4, 6, 12.
Prime and Composite Numbers	Primes have exactly two factors (1 and itself); composites have more. Example: 2, 3, 5 are prime; 4, 6, 8 are composite.
GCF and LCM	GCF: largest common factor; LCM: smallest common multiple. Use prime factorization. Example: GCF of 12 and 18 is 6; LCM is 36.

Table 4.1: Summary of Key Concepts

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